

CS 421 --- Type Semantics Activity (Polytype Version) --- Solutions

Proofs

Time estimate: 25 minutes.

Create proofs for the following judgements according to the given rules.

Problem 4) $\{id: \forall \alpha. \alpha \rightarrow \alpha, y: Int\} \vdash (id\ y) : Int$

$$\frac{\frac{\frac{}{\{id: \forall \alpha. \alpha \rightarrow \alpha, y: Int\} \vdash id : \forall \alpha. \alpha \rightarrow \alpha} \text{VAR}}{\{id: \forall \alpha. \alpha \rightarrow \alpha, y: Int\} \vdash id : Int \rightarrow Int} \text{INST} \quad \frac{}{\{id: \forall \alpha. \alpha \rightarrow \alpha, y: Int\} \vdash y : Int} \text{VAR}}{\{id: \forall \alpha. \alpha \rightarrow \alpha, y: Int\} \vdash (id\ y) : Int} \text{APP}$$

Problem 5) $\{y: Int, z: String\} \vdash (\lambda f. (f\ y, f\ z)) (\lambda x. x) : (Int, String)$

$$\frac{\frac{\frac{}{\Gamma \vdash x : \alpha} \text{VAR}}{\Gamma \vdash \lambda x. x : \alpha \rightarrow \alpha} \text{ABS} \quad \frac{\text{No way to use } f \text{ here...}}{\Gamma \cup \{f : \alpha \rightarrow \alpha\} \vdash (f\ y, f\ z) : (Int, String)} \text{PAIR}}{\Gamma = \{y : Int, z : String\} \vdash (\lambda f. (f\ y, f\ z)) (\lambda x. x) : (Int, String)} \text{APP}$$

Problem 6) $\{x: Int, y: String\} \vdash \text{let } f = \lambda x. x \text{ in } (f\ x, f\ y) : (Int, String)$

$$\frac{\frac{\frac{}{\Gamma \vdash x : \alpha} \text{VAR}}{\Gamma \vdash \lambda x. x : \alpha \rightarrow \alpha} \text{ABS} \quad \frac{\frac{\frac{}{\Gamma' \vdash f : \forall \alpha. \alpha \rightarrow \alpha} \text{VAR}}{\Gamma' \vdash f : Int \rightarrow Int} \text{INST} \quad \frac{\frac{}{\Gamma' \vdash x : Int} \text{VAR}}{\Gamma' \vdash f\ x : Int} \text{APP} \quad \frac{\frac{\frac{}{\Gamma' \vdash f : \forall \alpha. \alpha \rightarrow \alpha} \text{VAR}}{\Gamma' \vdash f : String \rightarrow String} \text{INST} \quad \frac{\frac{}{\Gamma' \vdash y : String} \text{VAR}}{\Gamma' \vdash f\ y : String} \text{APP}}{\Gamma' \vdash f\ x : Int \quad \Gamma' \vdash f\ y : String} \text{PAIR}}{\Gamma' = \{\forall \alpha. \alpha \rightarrow \alpha\} \cup \Gamma \vdash (f\ x, f\ y) : (Int, String)} \text{LET}}{\Gamma = \{x : Int, y : String\} \vdash \text{let } f = \lambda x. x \text{ in } (f\ x, f\ y) : (Int, String)} \text{GEN}$$